GRADUATE COURSE · ATMOSPHERIC GENERAL CIRCULATION

The Atmospheric General Circulation

Introduction — what this course is about, and how the textbook is organized

Textbook: J. M. Wallace, D. S. Battisti, D. W. J. Thompson & D. L. Hartmann, *The Atmospheric General Circulation* (Cambridge University Press)

Based on the book's Preface

Understanding the macroscopic behavior of the global circulation

The global circulation, viewed macroscopically

- Its structure and evolution
- How it mixes and disperses trace constituents
- How it mediates radiative transfer, cloud microphysics, and microscale turbulence
- How it interacts with the oceans, the biosphere, and the cryosphere
- How it imparts a regional character to climate change

From specialty to mainstream. When first introduced ~70 years ago, general circulation courses were specialty courses; today they connect atmospheric dynamics with the other subdisciplines and elucidate the behavior of the global atmosphere as a system.

An analogy from physics

General circulation : atmospheric science
≈ Statistical mechanics : physics

Interpreting the macroscopic behavior of the global circulation in terms of the dynamics and statistical properties of **ensembles of waves and vortices** of which it is made up.

Synoptic meteorology → the entities that shape and mediate day-to-day **weather**
General circulation → the entities that shape and mediate **climate**

WHAT THE TERM CONNOTES

Many kinds of weather systems on many space and time scales

The book's cover image is a single, typical frame from an animation — global in scope, ranging in scale from ~10 km to planetary. The circulation is rendered by the clouds it produces; the oceans, cryosphere, and terrestrial biosphere with which it interacts are also visible.

~10 km

convective clouds, microscale turbulence

~100–1000 km

cyclones, easterly waves, fronts

~1000–10,000 km

jet streams, storm tracks, stationary waves

Planetary

Hadley cell, ITCZ, annular modes, MJO

space scale →

Interacting with: Oceans · Cryosphere · Biosphere

Sixty years of a course, distilled into a monograph

1962–66

JMW a graduate student at MIT; Victor Starr's general circulation course

early 1970s

JMW begins teaching the graduate course at Univ. of Washington, closely paralleling Starr's

1970s–80s

New material: stratosphere (with James Holton); zonally varying extratropical circulation; wave–mean flow interaction breakthroughs

~10 years ago

Chih-Pei Chang (Holton's early student) urges JMW to publish the notes as a monograph

It took a team. Surveying the voluminous literature and organizing it into a coherent narrative with illustrations and a supporting mathematical framework required more than one author.

DSB — orderly development of the governing equations with scaling considerations · **DWJT** — Chapters 8 and 2, graphics design · **DLH** — global maps and meridional cross sections; Section 2.4

THE SCOPE

What this book is — and what it is not

Summarizes the major advances of the past ~70 years and offers a brief history of how they came about — emphasizing the achievements of the broader community over individual contributions.

vs. early general circulation textbooks

- Less focused on how the atmosphere fulfills the balance requirements (mass, angular momentum, energy)
- A more holistic treatment of the underlying dynamics
- Explains phenomena on a fundamental level, starting from the equations of motion

vs. other dynamics textbooks

- Oriented toward interpretation and explanation of phenomena, less toward applying general principles
- For the visual learner: ~1000 panels of graphics, far exceeding the number of equations
- Touches on a wide range of phenomena, many not found in other textbooks

vs. climate dynamics & planetary atmospheres

- Focus on the general circulation as it exists today — not the distant past, nor multidecadal change
- More focused on the circulation and on departures from the globally averaged temperature profile than on the profile itself

THE CONTENT

Roadmap: six parts, 21 chapters, Appendices A–F

Part I — Ch. 1–2

Background

Observations and models; heuristic models — heat engine, rotation, gravity waves

Part II — Ch. 3–6

Balance requirements

Angular momentum, trace constituents & water vapor, total energy, mechanical energy cycle

Part III — Ch. 7–8

Dynamics of the zonal mean flow

Axisymmetric vortex, wave–mean flow interaction, TEM

Part IV — Ch. 9–10

Stratospheric general circulation

Brewer–Dobson, sudden warmings, QBO, equatorial waves

Part V — Ch. 11–13

Zonally varying extratropical tropospheric circulation

NH winter climatology, stationary waves, transients

Part VI — Ch. 14–21

The tropical general circulation

ITCZ, convection, seasons, ENSO, MJO, waves, TCs, tides

Zonally averaged — the atmosphere treated as if Earth were an aquaplanet

Zonally varying — longitudinal structure of climate and variability

Background: observations, models, and an intuitive feel for the dynamics

Chapter 1 — Atmospheric Observations and Models

- Development of the observing system from the mid-20th century onward
- A “mini-atlas”: jet streams, Hadley cell, storm tracks, intertropical convergence zones, …
- Simplified equation sets via the hydrostatic and geostrophic approximations
- A brief history of general circulation modeling

Chapter 2 — Heuristic Models of the General Circulation

- Radiative–convective equilibrium; steady motions driven by heating gradients; “Tropic World”
- The influence of planetary rotation (a “spin-up” experiment, varying Ω), frictional drag, and gravity waves — the hidden messengers
- The general circulation as a thermodynamic cycle of reversible processes — a heat engine

Appendix A

The formalism for decomposing the flow:

time mean + transients
zonal mean + eddies (waves)

These decompositions underlie every part of the book — and the analyses you will do yourselves in this course.

Balance requirements: Starr's MIT course, seen with today's observations

Four conservation constraints

- Ch. 3 angular momentum · Ch. 4 mass balance of trace constituents (carbon, water vapor, PV) · Ch. 5 total energy · Ch. 6 the mechanical energy cycle
- Easier to cover now: no extensive error analysis or elaborate justifications for neglecting small terms

Observations Starr could only dream of

- Precise length-of-day and top-of-atmosphere net radiation measurements
- Reliable geographical distributions of precipitation and surface energy fluxes
- Global satellite fields of water vapor and a host of other trace substances

Decomposition of the horizontal wind field

- Axially symmetric part: zonally averaged zonal wind + mean meridional circulations
- Eddy (wave) part: departure from the zonal mean — transports mass, zonal momentum, and energy meridionally, like down-gradient eddy diffusion

Eddy transports

Represented by zonally and time-averaged **covariance terms** (e.g. $[u^*v^*]$, $[v'T']$) that appear in the conservation equations.

Derived in Appendices A and B.

Dynamics of the zonal mean flow: beyond the balance requirements

Chapter 7 — Dynamics of the Zonal Mean Flow

- Transcends the balance-requirement approach, in which angular momentum and energy are self-contained topics
- Treats the general circulation holistically via the equations governing the zonally symmetric flow
- Rooted in Eliassen (1951) on the axisymmetric vortex — the topic Charney taught in his MIT Planetary Circulations course; presented here in simplified form

Chapter 8 — Wave–mean flow interaction

- Charney lamented that he did not know how to represent eddy transports once waves reach finite amplitude
- Late 1970s: the transformed Eulerian mean (TEM) formalism — not just “what maintains the trade winds and westerlies?” but why they exist
- Nonlinear life cycle of baroclinic waves in a primitive-equation model as a concrete illustration of two-way interaction
- Brief look at the annular modes

Conceptually more advanced

Requires more mathematical manipulation than Part II.

The payoff: an explanation of the two-way wave–mean flow interaction that had previously eluded general circulation theorists.

The stratospheric general circulation

Chapter 9 — The Global Stratospheric Circulation

- The global, time-varying Brewer–Dobson circulation
- Stratosphere–troposphere exchange
- Planetary waves in the winter stratosphere; sudden warmings
- Common thread: breaking of Rossby waves dispersing upward from below lets air parcels flow poleward under the constraint of angular momentum conservation

Chapter 10 — Wave–Mean Flow Interaction in the Tropical Stratosphere

- The quasi-biennial oscillation (QBO) in zonally averaged zonal wind, driven by breaking inertio-gravity waves dispersing upward from directly below
- The family of equatorially trapped planetary waves and the linear shallow-water theory that accounts for them
- Two-sided wavenumber–frequency spectra vs. theoretical dispersion curves

Linear theory meets data

The comparison of observed wavenumber–frequency spectra with theoretically derived dispersion curves is a graphic example of how **linear theory can be of use in interpreting observations**.

Wave breaking is the recurring mechanism in both chapters.

The zonally varying extratropical tropospheric circulation

A change of viewpoint

- Up to Part IV the atmosphere is treated as if Earth were an aquaplanet — no significant longitudinal variations in climate
- Coverage limited, for brevity, to the Northern Hemisphere wintertime circulation

Chapter 11 — The NH Winter Zonally Varying Climatology

- Jet streams, stationary waves, subpolar lows, and subtropical highs
- Numerical experiments with a state-of-the-art GCM isolate the roles of orography, zonally varying diabatic heating, and tropical forcing
- Including nonlinear interactions with the zonally averaged zonal wind

Chapters 12–13 — High and Low Frequency Extratropical Transients

- Ch. 12: periods shorter than ~1 week · Ch. 13: periods longer than ~1 week
- Defined on the basis of time-filtered data

Why it matters

Longitudinal structure is what gives climate its regional character — and a zonal mean alone cannot describe it.

The tools: time-mean vs. transient decomposition and time filtering (Appendix A, C).

The tropical general circulation

Year-round structure and energetics

- Ch. 14 (annual mean circulation): the ITCZ/cold tongue complexes, equatorial stationary waves, what determines the rain rate climatology
- Ch. 15 (tropical convection): the energy balance, WTG balance, convectively coupled waves, self-aggregation

Low-frequency, planetary-scale variability

- Ch. 16 climatological mean annual cycle · Ch. 17 ENSO · Ch. 18 Madden–Julian oscillation

Higher-frequency and regional phenomena

- Ch. 19 (day-to-day variability): equatorially trapped waves, easterly waves, extratropical influences
- Ch. 20: warm core tropical vortices — tropical cyclone structure, genesis, and modulation
- Ch. 21: diurnal and higher frequency variability — tides, external modes, gravity waves

Documenting and interpreting

Each chapter documents the observed structure or variability, then interprets it — from the year-round ITCZ to sub-daily tides.

Chapter 21 ties together the sections on sub-daily inertio-gravity waves scattered through the book.

Figures, exercises, online tools, and data

Figures

Most numbered figures carry titles; captions emphasize what is shown rather than how it was produced. Data sources and methods are in Appendix F; dataset URLs on the companion web page.

Exercises

Follow most sections. Some have well-defined answers; those marked with an asterisk (*) are open-ended — proposed topics for class discussion, projects, or term papers.

Solutions Manual & online tools

Introduces online tools to access gridded atmospheric and oceanic fields, display them, and perform analyses — time and zonal averaging, compositing, EOF analysis.

Data behind the figures

Mostly ERA-Interim (discontinued 2019); gravity-wave and many Part VI figures use ERA5. Unless noted, ERA-Interim figures are not appreciably different from ERA5 counterparts.

Appendices A averaging operations · B zonal momentum balance · C teleconnection methods (EOF, MCA) · D weak temperature gradient scaling · E symbols & abbreviations · F extended figure captions. Online: high-resolution PDFs, extended mini-atlas, animations library.

In this course we will follow the book's “hands-on” spirit: analysis code for reproducing and exploring what we learn each week will be shared through the course GitHub page.

THE COMMUNITY

A research community dating back to the late 1940s

Pioneers (late 1940s–)

- Carl-Gustaf Rossby
- Victor Starr
- Jule Charney
- Eric Eady
- Edward Lorenz
- Arnt Eliassen
- Alan Brewer

Arriving in the 1960s

- Syukuro Manabe
- Taroh Matsuno
- James Holton

In the early years the field was populated by their former students and postdocs, many of whom made important contributions in their own right.

A community still growing — and still changing. Progressively more diverse in nationalities and institutions; in recent years many more women hold faculty and research positions. Sustained efforts are needed to attract more minorities into the geosciences: better K–12 STEM education alone is not enough — students need role models who convince them, from their early years, that they can enjoy and master science and math. Some current students contribute through departmental outreach to K–12 schools.

LOOKING AHEAD

Three questions to keep in mind all semester

1. What are the “building blocks” of the general circulation, and what do their statistics (means, variances, covariances) look like?
2. What balance requirements must the circulation satisfy, and which eddy transports fulfill them?
3. Why do these features exist at all — how do waves and the mean flow shape one another?

→ **Next lecture — Chapter 1.** The observing system since the mid-20th century, the “mini-atlas” of building blocks, and the time-mean / zonal-mean decomposition of Appendix A.